

Does GRPO teach a small model to reason over molecules — or just elicit what it already knows?

A symbolic RDKit verifier as reward on a 0.5B model, read through **pass@k**: we separate RL **sharpening** (same ceiling) from RL **expanding** capability — and find a sharp, reward-driven failure mode.

Model Qwen2.5-0.5B-Instruct

Reward RDKit verifier (+ shaped partial credit)

Tasks MolecularIQ · held-out trainPool split

Eval pass@k · k≤256 · T=0.8

Train GRPO from instruct · 1xA100

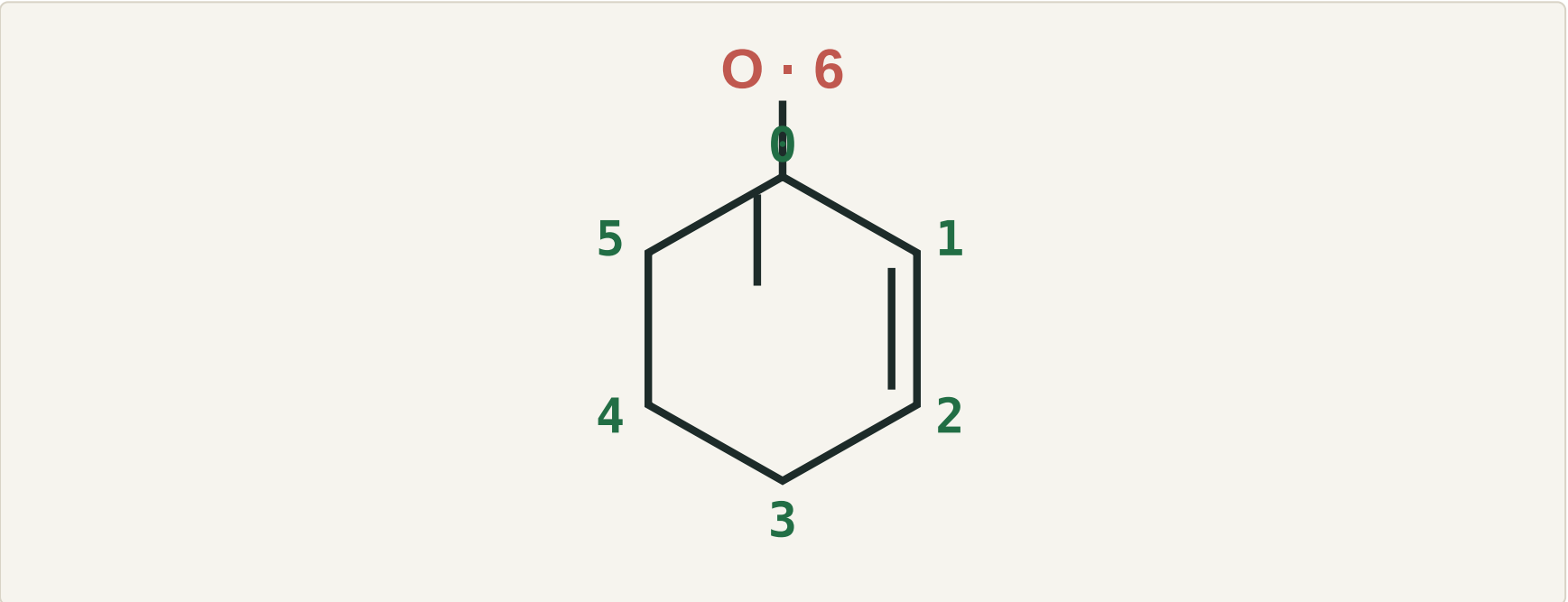
BACKGROUND

Structure understanding, verifiably

Chemistry LLMs are scored on benchmarks that leak into pretraining. **MolecularIQ** instead asks only **symbolically verifiable** questions about the molecular graph — every answer is checked *exactly* with RDKit, isolating reasoning from memorization and giving a clean, non-gameable RL reward.

- counting
- indexing
- constrained generation

WHAT THE VERIFIER CHECKS



TASK Return the 0-based indices, in SMILES order, of every atom in a ring.

ANSWER `<answer>{"ring_index": [0,1,2,3,4,5]}</answer>`

- ✓ RDKit recomputes the exact set and grants reward
- 1. only on a full match — no LLM judge, no leakage. (Atom 6, the O, must be excluded.)

HYPOTHESES

H1 · TEACH VS. ELICIT (MAIN)

pass@k separates capability from sharpening. If GRPO raises pass@1 but the **base catches up at large k**, RL only *elicited* existing skill; if GRPO stays **above base at all k**, it *expanded* capability.

H2 · COMPOSITIONAL TRANSFER

Skills trained on single properties should transfer to never-trained **compositions** (single → multi) and from **analysis** → **synthesis** (counting → generating to a target).

H3 · REWARD REACHABILITY

GRPO improves a task only where **exact** verifier success is reachable. Where it is not, GRPO optimizes the dense **partial-credit proxy** instead — and can drop **below** the base.

SETUP

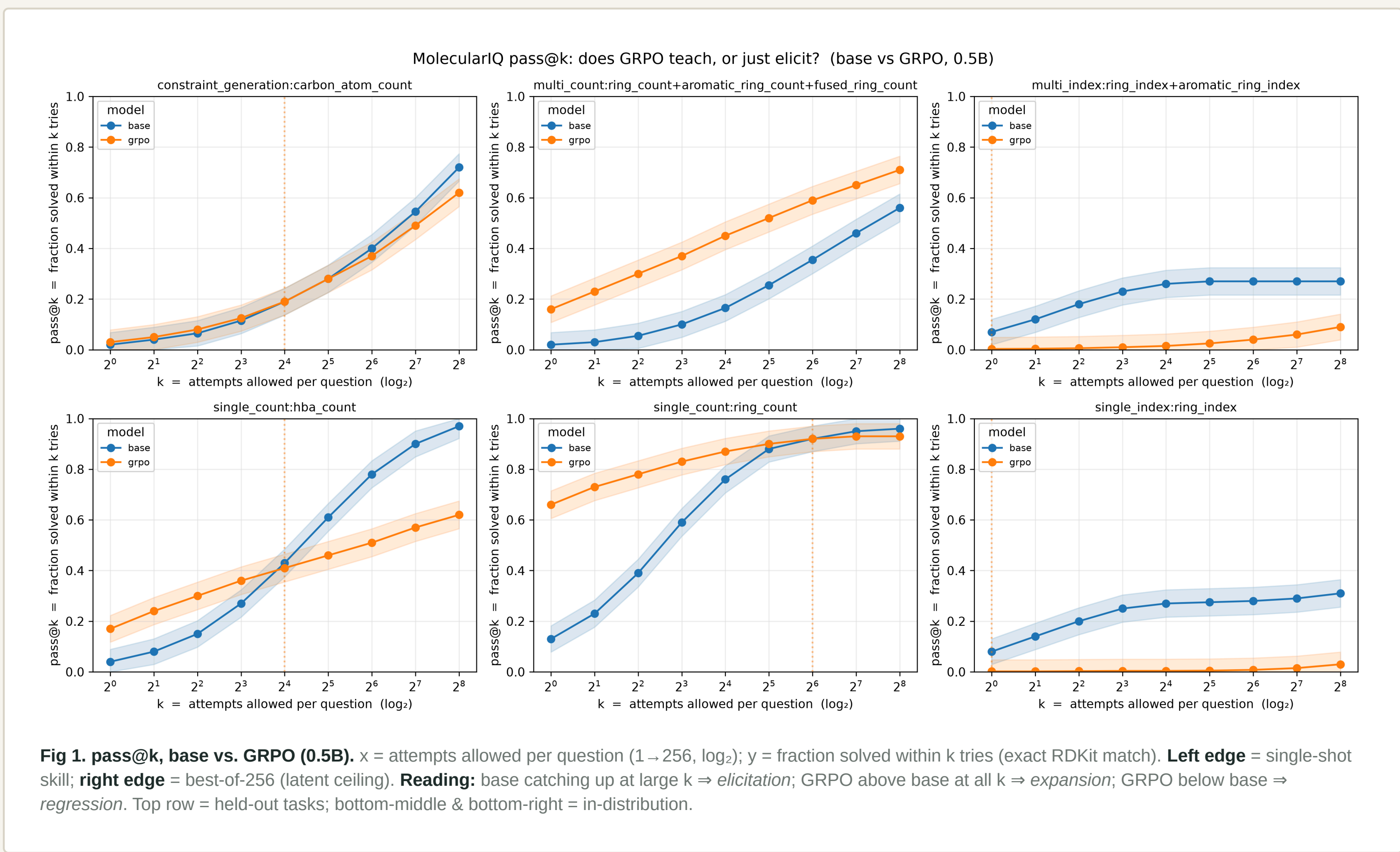
- base** — Qwen2.5-0.5B-Instruct, no fine-tuning (reference).
- GRPO** — full-parameter GRPO from the instruct base; verifier + shaped reward.
- Train tasks** — pooled single-property counting + indexing.
- Held out** — multi-property compositions, constrained generation, an unseen property (HBA).

Eval = disjoint tail of the MolecularIQ trainPool; questions & gold from the official generator, scored by the official RDKit verifier (not the leaderboard partition). pass@k from 256 samples (Chen et al. estimator).

HEADLINE RESULT

GRPO elicits, it doesn't teach — and hacks the reward where it can't reach.

Verifier-GRPO boosts **first-try** counting accuracy and transfers compositionally to unseen **multi-count** — but the base model's pass@k **catches up** at large k (sharpening, not new capability). On atom-**indexing**, where exact success is unreachable, GRPO over-covers to farm partial credit and falls **below base**.



Making indexing learnable — why it failed → how to fix

Indexing needs exact symbolic execution (parse SMILES → number heavy atoms → perceive rings), so the base almost never samples a correct set and RL has nothing to reinforce. Concretely:

- **Cold-start SFT on RDKit-generated reasoning traces, then GRPO** — RL can't reinforce a skill the base never samples.
- **Difficulty curriculum** — tiny molecules first, so the exact reward becomes reachable.
- **Remove the over-coverage incentive** — precision/size penalty, or binary-only reward above a competence threshold.
- **Tool use** — let the model call RDKit and offload the symbolic work; makes model size nearly irrelevant for this task family.

Scale (a bigger base model) helps too, but is the brute-force version of cold-start.

THREE REGIMES, ONE MECHANISM

● Elicitation (in-distribution)

single_count : ring_count. pass@1 **0.13** → **0.66**, but base overtakes by **k=64**. RL sharpened the base distribution — same ceiling. (Reproduces Yue et al. 2025 in chemistry.)

● Expansion + transfer (held-out)

multi_count. GRPO above base at **every k** (0.16 vs 0.02 @k=1; 0.71 vs 0.56 @k=256). Single-property skills **recombined** into an unseen task = real new capability.

● Reward-hacking / regression

single_index, multi_index. GRPO ≈ 0, **below base** (~0.3). Exact match unreachable → only the dense proxy gives signal → it gets hacked.

REWARD-HACKING, CAUGHT IN THE ACT

TASK: RING_INDEX · GREEDY ACC = 0.00 (VS BASE ~0.31)

```
gold (one molecule):
{"ring_index": [5,6,7,8,9,10]}

GRPO emits wide contiguous ranges (distinct=0.40 — not collapse):
[3,4,...,12] [0,1,...,8] [6,7,...,19]
```

→ over-covers to farm Jaccard overlap (high recall, low precision); the partial-credit signal is misaligned with exact-match, so it **never** matches.

TAKEAWAYS

What this shows

- pass@k cleanly separates RL **sharpening** from **new capability** — a 0.5B + exact verifier is an ideal, cheap testbed for the elicitation-vs-expansion debate.
- GRPO elicits in-distribution and transfers **compositionally** where the exact reward is reachable (counting).
- Where exact success is unreachable (indexing), dense partial credit gets **hacked** and GRPO regresses below base — a concrete RLVR failure mode *with* its mechanism.

base GRPO